

What the sound descriptors measure

Level, spectral shape, and change—with the exact weighting and silence conventions.

02.01 · Level and spectral balance

EXPLANATION

RMS measures amplitude without cancellation between positive and negative samples. sllr computes it on the unwindowed frame. It is neither sound-pressure level nor a complete model of perceived loudness.

Centroid measures the spectrum's balance point and spread measures its width around that point. Both use magnitude weights, including the DC bin. A bright sound often has a high centroid, but the interpretation depends on the recording and listening context.

A useful experiment is to multiply the same floating-point recording by a nonzero gain: RMS should scale while centroid and spread should stay constant, away from silence thresholds and numerical floors.

MATHS

With raw frame samples $x_m[n]$ and magnitude weights $a_m[k] = A_m[k] / \sum_j A_m[j]$,

$$\text{RMS}_m = \sqrt{\frac{1}{N} \sum_n x_m[n]^2},$$

$$c_m = \sum_k f_k a_m[k], \quad s_m = \sqrt{\sum_k (f_k - c_m)^2 a_m[k]}.$$

For $y[n] = gx[n]$ with $g \neq 0$,

$$\text{RMS}(y) = |g| \text{RMS}(x), \quad c(y) = c(x).$$

This prediction assumes no clipping. The in-app experiment operates in floating point without playback clipping.

02.02 · Distribution of spectral power

EXPLANATION

Entropy describes how evenly power is distributed across bins. Flatness compares the geometric and arithmetic means of power. Roll-off finds the first frequency bin containing 85% of cumulative power.

These descriptors answer different questions. Entropy is not a musical-complexity score, flatness is not a categorical noise detector, and roll-off is a chosen descriptive threshold.

The implementation returns zero for silence-derived spectral descriptors. Those zeros are placeholders, not evidence that silence has a meaningful brightness or spectral distribution.

MATHS

Let $K = N/2 + 1$ and $p_k = P[k] / \sum_j P[j]$. Normalized entropy is

$$E = -\frac{\sum_k p_k \ln p_k}{\ln K}, \quad 0 \ln 0 = 0.$$

With $\epsilon = 10^{-24}$, flatness is

$$F = \frac{\exp(K^{-1} \sum_k \ln \max(P[k], \epsilon))}{K^{-1} \sum_k P[k]}.$$

The result is clamped to $[0, 1]$. Roll-off is

$$r = f_{k_*}, \quad k_* = \min \left\{ k : \sum_{j=0}^k P[j] \geq 0.85 \sum_j P[j] \right\}.$$

The code treats $\sum_k P[k] < 10^{-24}$ as silent. Floors matter for very quiet signals.

02.03 · Changes and crossings

EXPLANATION

Spectral flux compares successive normalized magnitude spectra and keeps only positive changes. It often responds to attacks, but silence-to-sound transitions also create peaks. The first frame has zero flux because it has no predecessor.

Zero-crossing rate counts changes between the nonnegative and negative sides of the waveform. It is sensitive to DC offset and noise; it is not a reliable pitch detector by itself.

Flux is normalized spectral change, so its units differ from RMS. Always read the units before comparing descriptor plots.

MATHS

For normalized magnitudes $a_m[k]$,

$$\text{Flux}_m = \sqrt{\sum_k \max(0, a_m[k] - a_{m-1}[k])^2}.$$

A spectrum whose magnitude sum is below the normalization threshold uses zero weights. Zero-crossing rate on the raw frame is

$$Z_m = \frac{1}{N-1} \sum_{n=1}^{N-1} |b_n - b_{n-1}|,$$

Here $b_n = \mathbf{1}[x_m[n] \geq 0]$. Zero belongs to the nonnegative side. The result is crossings per sample rather than crossings per second.